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Pr. Brahim Hamid (IRIT Laboratory, UT2J)
Dr. Marcos Didonet Del Fabro (CEA List)

Subject: Application for PhD Position – AI-Assisted Security Engineering for Digital Twins (PC3)

Dear Pr. Hamid and Dr. Didonet Del Fabro,

I am writing to express my profound interest in the PhD position on "AI-Assisted Model-Driven Security Engineering for Digital Twins." With my robust engineering background in Artificial Intelligence, Graph Neural Networks (GNNs), and automated cybersecurity architectures, I am highly motivated to tackle the challenge of developing an automated round-trip framework for the Asset Administration Shell (AAS) specification.

In my engineering of the "Autonomous Cyber-Shield Platform," I confronted the complexities of closing the loop between threat mitigation and system architecture. I designed a highly deployable architecture that integrated a Reinforcement Learning penetration tester directly against an autonomous firewall defender (Blue AI). By utilizing Adversarial Co-Evolution, I modeled complex security environments where the system autonomously learned and fed security rules back into the baseline architecture. This hands-on experience in AI-driven cybersecurity perfectly aligns with your objective to integrate AI assistance into the modeling of Digital Twins. I am fully prepared to extend these capabilities into the AAS specification, ensuring that model-driven engineering processes can seamlessly adapt to dynamic security threats in real-time.

Furthermore, my deep knowledge of Model-Driven Engineering positions me to design the necessary formalisms to bridge the gap between design-time security and run-time monitoring. I am eager to leverage my skills to develop the required Proof of Concept framework and contribute meaningfully to the EDT case studies. Please find attached my resume and academic transcripts. I look forward to discussing how my expertise can accelerate the development of an automated security framework within your teams.

Sincerely,

Ismail Aissaoui